



WILLON SMITH

AI Engineer | Technical Delivery Lead

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PROFESSIONAL SUMMARY

AI Engineer and Technical Delivery Lead with 5+ years turning business needs into production software, currently leading delivery for a UK AI consultancy. Named lead on two client accounts, owning intake, scoping, build, deployment and production support; work spans full-stack platforms, AI agents, analytics warehouses and automation pipelines built with Claude, Python and TypeScript. Works evidence-first: CI-enforced test coverage, automated evaluation gates on AI releases, and a security baseline on every client-facing surface. Experienced remote collaborator with UK teams (SAST, GMT+2, strong UK-hours overlap).

TECHNICAL SKILLS

AI & Automation	Claude, Claude Code, Anthropic API, AI agents, LLM tool use, prompt caching, RAG pipelines, embeddings, LLM routing and cost control, AI evaluation testing, n8n, workflow orchestration
Development	Python, TypeScript, JavaScript, React, Django, Django REST Framework, FastAPI, Flask, Node.js, SQL, HTML, CSS, Google Apps Script, REST APIs, GraphQL, Git, GitHub
Data & Cloud	PostgreSQL, DuckDB, dbt, MongoDB Atlas, AWS ECS, AWS RDS, AWS S3, Terraform, Docker, GitHub Actions, Vercel, Neon, Render, Hetzner
Platforms & Integrations	Microsoft Graph, Outlook automation, Business Central OData, Monday.com, Airtable, Linear, Twilio, WhatsApp API, Telegram Bot API, Calendly, Google Workspace, AppSheet
Infrastructure & Security	Linux server hardening, UFW, fail2ban, NGINX, TLS, Tailscale, Grafana, Loki, OAuth 2.0, RBAC design, API spend controls, systemd
Delivery & Documentation	Requirements gathering, MVP scoping, sprint cycles, stakeholder management, client-facing reporting, technical documentation, training delivery

PROFESSIONAL EXPERIENCE

AI Engineer & Technical Delivery Lead

Agency AI (UK AI consultancy) | Remote | Fixed-term contract

April 2026 – Present

- Own two client accounts end to end, covering intake, scoping, build, deployment and production support, delivering AI systems, automation and production engineering across seven UK client businesses in construction, distribution, brand activation, recruitment and consumer software
- Sole engineer on a business operating platform for a Scottish construction group (Django, React/TypeScript, PostgreSQL): approval-gated process engine, four-layer RBAC and a dynamic form builder replicating the client’s real contract paperwork. Live in production with 260+ backend tests and a CI-enforced 85% coverage gate
- Surfaced £1,726/week of live plant-hire spend and roughly £600/week of forgotten off-hire charges on the first sync of real data, replacing manual spreadsheet tracking with database-backed, phone-editable trackers and an executive analytics view
- Built a permission-aware AI assistant (Claude tool-calling over a guarded read-only SQL layer, write actions requiring user confirmation), live across three channels: in-app, Telegram and WhatsApp; adversarial testing I ran found and closed two data-leak paths
- Delivered an asset-analytics platform for a UK brand-activation company serving global drinks brands: dbt warehouse, live Vercel portal within ten days of confirmation, and an AI query agent gated by a 15-case automated evaluation suite, then rebuilt it for a second brand’s tender in under a week, every figure verified against live SQL

- Owned production support of a live n8n recruitment pipeline: built error alerting and heartbeat monitoring after an 18-hour silent outage, retry-hardened ~50 nodes across 7 workflows, and re-architected fragile email triggers to scheduled polling with duplicate detection, eliminating recurring stalls
- Fixed production faults on a document-RAG web application (FastAPI, MongoDB Atlas): moved blocking document parsing to background tasks with batched embeddings. Uploads that hung 5+ minutes returned in under 3 seconds, verified in production on the exact files users reported
- Re-architected a client's live email-campaign automation into a deterministic four-workflow system (n8n, Microsoft Graph) with code-enforced guardrails (follow-up caps, circuit breaker, per-recipient cooldown), passing a 180-message concurrency regression with zero wrong-name errors before go-live
- Operated the consultancy's engineering standards: dedicated hardened Linux servers per client with Tailscale-only admin access and Grafana/Loki observability, plus a 10-point security baseline with a documented kill switch. Set its three-tier LLM routing and prompt-caching standard after a single-model agent burned ~\$50/day

Workflow Automation & Application Development Lead

EZEMind AI | Bloemfontein (Remote)

March 2025 – March 2026

- Led the design and delivery of end-to-end automation solutions for SME clients, translating business requirements into structured technical deliverables and production-ready workflows
- Architected and deployed complex n8n automation workflows that streamlined the complete payment cycle for multiple clients, reducing manual processing time by over 60%
- Developed EzeFlow, a custom business application with integrated paycycle management and auditor modules, enabling automated financial tracking and compliance reporting
- Built AVA, an AI virtual assistant on Claude, automating routine business operations for SME clients
- Engineered Google Workspace integrations connecting n8n with Google Sheets and Apps Script for automated reporting and complex calculations, and built custom AppSheet business applications
- Configured Monday.com CRM for multiple clients covering sales pipelines and automations, delivered corporate training programmes and documentation, and built and launched the company website

IT Solutions Consultant & Freelance Developer

Self-Employed | Bloemfontein

July 2024 – March 2025

- Provided end-to-end IT consulting for SME clients including website development, email configuration, application deployment and system administration, creating and maintaining responsive client websites
- Managed client onboarding processes, gathering requirements and delivering tailored technical solutions
- Delivered online English lessons to 200+ students (TEFL certified, 120-hour Universal TEFL Certificate)

IT Technician & Technical Analyst

E2E Group / WRCON | Bloemfontein

February 2023 – February 2024

- Provided comprehensive IT support across the organisation including system setup, troubleshooting, OneDrive administration, performance optimisation, and user account management
- Configured and managed Geographic Information Systems (GIS) using QGIS for municipal asset mapping and analysis
- Conducted condition assessments and completeness testing for municipal asset inventories, generating auditor-grade reports and performing data integrity and duplicate detection across large asset databases
- Developed and launched the company website, handling frontend development (HTML/CSS) with PHP backend integration

Junior Developer & Workflow Specialist

National Real Estate | Bloemfontein

December 2021 – February 2023

- Designed, built and maintained DocuWare document management workflows, merge forms and web forms tailored to business requirements, and implemented user role structures and access controls for system security
- Authored technical documentation and ran user training sessions, and managed database records in FixFlow including new employee onboarding and system access across platforms

Technical Support Specialist

Vodacom (Techzone) | Bloemfontein

April 2021 – November 2021

- Diagnosed and resolved hardware, software, network and GSM issues for consumer and business clients; configured ICT equipment and networks, managed data migration and backups, and coordinated warranty repairs and technical inventory

General Manager & Web Developer

Heaven Cloud | Bloemfontein

October 2018 – December 2020

- Led a small team delivering IT solutions and web development services to business clients, building frontend web solutions, providing remote client support, and managing client relationships and day-to-day operations

EDUCATION & CERTIFICATIONS

System Engineer Skills Certificate (MICT SETA accredited)

2017

ATTI – Advanced Technology Training Institute, Bloemfontein

Short courses: PC Technician, Networking Technician, Windows Server 2008 and 2012 (installation, configuration, administration, advanced services), Server Infrastructure Design and Implementation, Microsoft Office. Awarded Certificate of Achievement (Silver) in System Engineering, March 2017

120-Hour Universal TEFL Certificate

International TEFL certification for English language instruction

National Senior Certificate (Matric)

2013

Sentraal High School, Bloemfontein

KEY PROJECTS & ACHIEVEMENTS

- **Company Website Rebuild:** Designed and built the consultancy's 12-page marketing site in-house with SEO preservation: structured data on every page, ranking pages rebuilt at their exact URLs with byte-for-byte title matches. Saved a quoted £15,000+ agency fee
- **Large-Scale Data Ingestion:** Ran the monthly AWS refresh for a consumer app's ~5M-record image-matching database: filtered a 10.3 GB, 19.2M-record dataset and delivered an 80,000-record ingestion at 99.8% completion on ECS and RDS pgvector with Terraform-managed infrastructure, writing streaming tools sustaining ~8,000 records/second

ADDITIONAL INFORMATION

Languages: English (fluent), Afrikaans (native)

Work Setup: Fully equipped home office with reliable internet connection and backup power

Timezone: SAST (GMT+2), strong overlap with UK/European working hours

Availability: Available from October 2026, on completion of current UK contract; open to remote contract or permanent roles